From Black to White Boxes – Interpretable Regression with the TRUST™ Algorithm

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Bio

- Ph.D. in Statistics, University of Wisconsin-Madison (La Caixa Fellow)
- Former Risk Analyst, European Central Bank
  - Led early applications of predictive ML in my division (2017)
  - I probably wouldn't be here today without that experience!
- Began my coding journey at Carnegie Mellon University in 2015, and later developed 3 open-source R packages with >100K downloads
- Creator of the TRUST™ algorithm and trust-free Python package

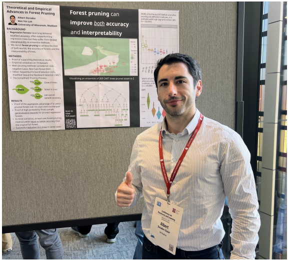


Figure 1: Presenting a recent paper at CPAL 2025 (Stanford University)

1. Review of Regression

- Main goal: to **estimate with enough accuracy** a **real-valued** outcome.
  - Examples: estimating **medical insurance charges** based on individual characteristics (later today), estimating a **disease marker** based on e.g. lifestyle choices, estimating **house prices** based on their attributes, estimating **demand for a good/service** based on certain variables
  - Formally: assuming $y = f(X) + \varepsilon$ (typically $\varepsilon \sim N(0, \sigma)$) we use data (X, y) to learn f from a set S of candidate models
- Other possible goals: **transparency**, fairness, efficiency
- Focus of this presentation: **accuracy + transparency = TRUST** (pun completely intended)
  - But **is that combination realistic** in practice? Not really – let's wrap up.

Just kidding... Although that was long-believed to be the case:

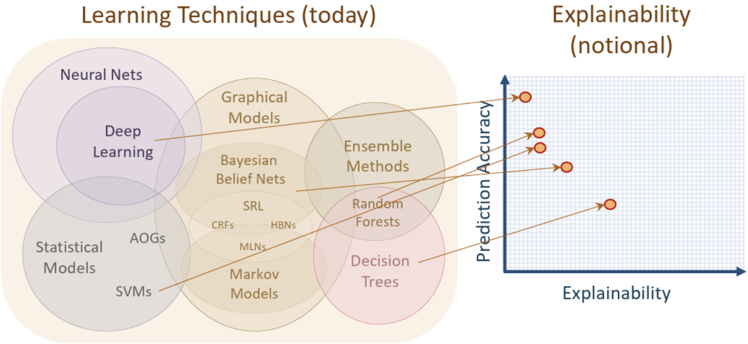


Figure 2: Source: DARPA's XAI Program (2016)

The reason for **hope**: multiple models (including, in some cases, *simple* ones) can describe the $X \rightarrow y$ map reasonably well (Rashomon Effect).

2. Interpretable Machine Learning Models

They include, in rough order of interpretability*:

- (Short) Decision Trees
- (Sparse) Linear Models and Generalized Linear Models (GLM's)
- (Sparse) Linear Model Trees
- (Sparse) Generalized Additive Models (GAM's) and friends (e.g. Splines, RuleFit, NodeHarvest)

*Context-dependent and global vs local notions

3. Introducing TRUST™ (PRICAI 2025)

- **T**ransparent: global and local explanations
- **R**obust: smart handling of missing values and extra/interpolation
- **U**ltra-**S**parse: relaxed Lasso leaves
- **T**rees: subgroup identification through axis-aligned partitions

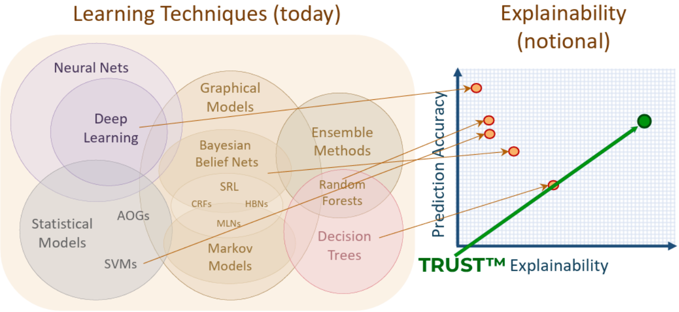


Figure 3: *Updated* DARPA's explainability landscape

4. Software Implementation

The TRUST algorithm is implemented in the Python package trust-free.

You can download it and install it for free from PyPI via:

pip install trust-free

and try for yourself all the functionality described in this presentation.

The latest version (2.1.4) is compatible with Windows, macOS, and Linux (including Google Colab and Kaggle).

Transparent

5.1 TRUST is Transparent

Globally

- Tree plot
- Variable importance scores
- ALE plots

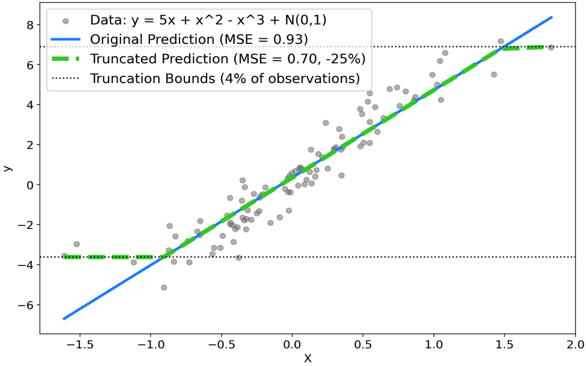
Locally

- Explain()
- Compare()

Robust

6.1 TRUST is Robust [extra/interpolation errors]

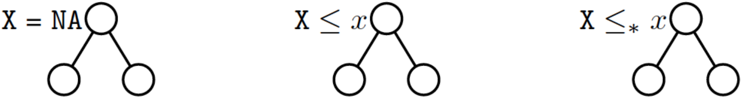
- What? Protect from extrapolation (and interpolation) errors
- Why? Reduce damage from predictions on instances outside training convex hull
- How? Prediction truncation, generalizing Loh et al., 2007



6.2 TRUST is Robust [missing values]

- What? Use information on missingness, then impute
- Why? The fact that a value is missing may itself be predictive
- How? Splits that consider missingness (Loh and Zhou, 2020):

3 types of splits on `X` if it has only 1 missing value code (`NA`):



"$X = \texttt{NA}$" means "Go left if X is missing"

"$X \leq x$" means " Go left if X is nonmissing AND $X \leq x$"

"$X \leq_* x$" means "Go left if X is missing OR $X \leq x$"

6.3 TRUST is Robust [high-dimensional prediction]

- What? Fit an elastic net at the root if $n \leq p + 1$
- Why? Have a (reasonable) unique solution and reduce overfitting. Very fast.
- How? Recall the elastic net problem given positive λ, δ

$$\min_{\beta} ||y - X\beta||_2^2 + \lambda||\beta||_1 + \delta||\beta||_2^2$$

Note: elastic net can be seen as a special case of a depth-0 TRUST tree.

Ultra - Sparse

7. TRUST is Ultra-Sparse

- Fit a relaxed Lasso model in the leaves
- Variable selection + regularization
- Relaxed Lasso, unlike Lasso, optimizes both aspects separately
- More efficient variable selection than stepwise or other heuristics
- Slightly more expensive than Lasso (2 hyperparameters to tune, $\lambda \geq 0$ and $\theta \in (0, 1]$)

$$\min_{\beta} \frac{1}{n} \sum_{i=1}^{n} (Y_i - X_i^T \beta \odot I\{k \in S_\lambda\})^2 + \theta\lambda ||\beta||_1$$

- $\theta = 1$ yields Lasso
- $\theta \to 0$ corresponds to Lasso-OLS hybrid (Efron et al., 2004)
- $\lambda = 0$ reverts back to OLS

8. Empirical Results (Summary)

Table 1: Mean unexplained variance and mean ranks by model across 60 datasets

Metric	TRUST	RF	M5'	CART	Lasso	NH
Mean	**0.33**	0.38	0.64	0.51	0.43	0.53
Std	0.03	0.03	0.04	0.03	0.04	0.03
Mean Rank	**1.70**	2.40	4.50	4.53	3.13	4.45
Std Rank	0.09	0.17	0.20	0.12	0.19	0.18

- TRUST is at least as accurate as Random Forest (RF)
  - TRUST has an edge under smaller samples (Tables 6-9 in PRICAI paper)
- TRUST is more accurate than any other interpretable model tested
- TRUST is much simpler than M5' (another linear model tree): 109 vs 17 parameters (median)

9. Example: Modeling Medical Insurance Charges

Curious to see **TRUST™** in action on a real-world dataset? Check out our Jupyter Notebook.